

C05-AT2-Short Answer Preparation

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QUESTIONS

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5 marks

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Q6
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Q1 Graph

5 marks

What do you call a graph where every vertex is reachable from every other vertex?

Your Answer

A connected graph is a type of graph in which there is a path between every pair of vertices. This means you can start at any vertex and travel to any other vertex by following the edges. If even one vertex cannot be reached from another, the graph is called disconnected.

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Q2 Graph's

5 marks

What structure is obtained when all cycles are removed from a graph?

Your Answer

Removing all cycles from a graph results in a structure called a Tree (if the graph is connected) or a Forest (if the graph is disconnected).
A cycle is a closed path that allows you to return to a starting node. When these are removed, every pair of nodes in a connected component is joined by exactly one unique path.
In short:
Connected graph without cycles = Tree

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Q3 Graph's

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What key property must a graph satisfy before applying topological sorting?

Your Answer

A graph must be a Directed Acyclic Graph (DAG) to apply topological sorting. That means the graph should have no cycles—only then you can line up the vertices in a valid order.

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Q4 Graph's

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Which traversal is used to find shortest paths in unweighted graphs?

Your Answer

Use BFS (Breadth First Search), because it explores neighbors level-by-level, so the first time you reach a node, you've found the shortest path in an unweighted graph.

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Q5 Graph's

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What is the time complexity of the Floyd-Warshall algorithm?

Your Answer

The time complexity of the floyd warshall algorithm is order of $O(v^3)$

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Q6Graph's5 marks

Q6 Graph's5 marks

Which property guarantees the correctness of an MST?

Your Answer

The correctness of a Minimum Spanning Tree (MST) is guaranteed by the Cut Property. This property states that for any cut of a graph, the edge with the minimum weight that crosses the cut must be part of the MST.

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Q7 Graph's

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Which algorithm is widely used in AI pathfinding?

Your Answer

A* (A-star) algorithm is widely used in AI pathfinding

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Q8 Graph's

5 marks

Which data structure is most efficient for implementing Prim's algorithm on a dense graph?

Your Answer

For a dense graph, the best data structure for Prim's algorithm is an Adjacency Matrix, because it allows checking edge weights quickly (since many edges exist).

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Q9 Graph's5 marks

What specific condition allows Dijkstra's algorithm to run faster than Bellman-Ford?

Your Answer

Dijkstra's algorithm is faster because it assumes all edge weights are non-negative. This allows it to use a greedy approach to find the shortest path more efficiently, whereas Bellman Ford must account for negative weights, making it more computationally expensive.

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Q10 Graph's

5 marks

Which shortest path algorithm is based on the principle of "Successive Approximation"?

Your Answer

The shortest path algorithm that works using the idea of "successive approximation" is the Bellman-Ford algorithm. It repeatedly relaxes edges, meaning it improves (approximates) the shortest distance step-by-step until the best result is obtained (and it can even detect negative cycles).

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Q11 Graph's

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Which algorithm is preferred for finding MST in a sparse graph?

Your Answer

Kruskal's Algorithm was the algorithm that was preferred for finding MST in a sparse graph

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Q12 Graph's

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Which algorithm is commonly used in GPS navigation systems?

Your Answer

GPS navigation systems commonly use the A* (A-star) algorithm for finding the shortest and fastest route based on road distance and estimated travel time.

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Q13 Graph's

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What happens to the MST if a constant value is added to every edge weight in the graph?

Your Answer

Adding a constant value C to every edge weight does not change the MST. Because the relative order of edge weights remains the same, Kruskal's and Prim's algorithms will select the exact same set of edges. The only change is the total weight of the MST, which increases by $C \times (V - 1)$, where V is the number of vertices.

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Q14 Graph's

5 marks

How is the "Relaxation" step mathematically defined in shortest path algorithms?

Your Answer

The relaxation step is defined by the following inequality: If the current known distance to a node (u) plus the weight of the edge connecting it to a neighbor (v) is less than the current known distance to that neighbor, then we update the neighbor's distance. Mathematically: if $\text{dist}[u] + \text{weight}(u, v) < \text{dist}[v]$, then $\text{dist}[v] = \text{dist}[u] + \text{weight}(u, v)$.

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Graph's

5 marks

Q15

Graph's

5 marks

In a DAG, which node will always have an in-degree of zero in a topological ordering?

Your Answer

In a DAG, the node that always appears first in a topological ordering is the node with in-degree = 0. Because it has no incoming edges, it can't be blocked by any prerequisite node, so it can always come first.

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Q3

Graph's

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Q4

Graph's

5 marks

✓

Q5

Graph's

5 marks

✓

Q6

Graph's

5 marks

✓

Q16 Graph's

5 marks

Which method for topological sorting uses recursion?

Your Answer

Topological sorting can be done using DFS (Depth First Search).
In DFS-based topological sort, we naturally use recursion to explore the nodes, then we push the node into the result after visiting its neighbors.
So, the method that uses recursion is DFS-based Topological Sorting.

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Q17 Graph's

5 marks

Which traversal approach is used in Kahn's algorithm?

Your Answer

Kahn's algorithm uses a BFS-like approach for topological sorting in a DAG. It repeatedly picks nodes with zero indegree, adds them to the order, and removes their outgoing edges. This reduces indegrees of neighboring nodes; whenever a node becomes zero indegree, it's added to a queue. The process continues until the queue is empty. If all nodes are processed, you get a valid topological order; if not, the graph has a cycle and topological sorting isn't possible.

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5 marks

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Q5

Graph's

5 marks

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Q6

Graph's

5 marks

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Q18 Graph's

5 marks

What is a graph without cycles called?

Your Answer

A graph that contains no cycles is called an acyclic graph. Because it lacks closed loops, it is impossible to return to a starting vertex by following a path of edges. Trees and Directed Acyclic Graphs (DAGs) are primary examples, often used in dependency modeling where topological sorting is required.

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Q19 Graph's

5 marks

Which graph representation uses a two-dimensional array?

Your Answer

A graph representation that uses a two-dimensional array is the Adjacency Matrix. It stores graph connections in a 2D table (matrix), where each cell tells whether an edge exists between two vertices.

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Q6
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5 marks

Q20 Graph's 5 marks

Which algorithm would you choose to find the shortest distance between all pairs of cities in a small network?

Your Answer

To find the shortest distance between all pairs of cities in a small network, The Floyd-Warshall algorithm was used. This algorithm is perfect for this task because it systematically checks all possible paths between every pair of nodes to find the absolute shortest route. It works by building up the shortest paths using intermediate cities, making it very reliable for small to medium-sized networks.

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